The Bucket

When we left the bucket, both it and the water within were spinning, with the water's surface forming a concave shape. The issue Newton raised is, *Why* does the water's surface take this shape? Well, because it's spinning, you say, and just as we feel pressed against the side of a car when it takes a sharp turn, the water gets pressed against the side of the bucket as it spins. And the only place for the pressed water to go is upward. This reasoning is sound, as far as it goes, but it misses the real intent of Newton's question. He wanted to know what it *means* to say that the water is spinning: spinning with respect to what? Newton was grappling with the very foundation of motion and was far from ready to accept that accelerated motion such as spinning—is somehow beyond the need for external comparisons.[1]

A natural suggestion is to use the bucket itself as the object of reference. As Newton argued, however, this fails. You see, at first when we let the bucket start to spin, there is definitely *relative* motion between the bucket and the water, because the water does not immediately move. Even so, the surface of the water stays flat. Then, a little later, when the water is spinning and there *isn't* relative motion between the bucket and the water, the surface of the water *is* concave. So, with the bucket as our object of reference, we get exactly the opposite of what we expect: when there is relative motion, the water's surface is flat; and when there is no relative motion, the surface is concave.

In fact, we can take Newton's bucket experiment one small step further. As the bucket continues to spin, the rope will twist again (in the other direction), causing the bucket to slow down and momentarily come to rest, while the water inside continues to spin. At this point, the relative motion between the water and the bucket is the *same* as it was near the very beginning of the experiment (except for the inconsequential difference of clockwise vs. counterclockwise motion), but the shape of the water's surface is *different* (previously being flat, now being concave); this shows conclusively that the relative motion cannot explain the surface's shape.

Having ruled out the bucket as a relevant reference for the motion of the water, Newton boldly took the next step. Imagine, he suggested,